# Evaluation Integrity in Multi-Agent Slice Admission Control: When Compliance Metrics Mask Training Collapse

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**Abstract**

Coordinating slice admission decisions across multiple gNBs is a genuinely multi-agent problem: does a shared, graph-structured cluster representation help per-gNB agents make better joint decisions than a single agent or independent learners? We study this with GAT-CTDE, a centralised-training, decentralised-execution architecture built on a shared Graph Attention encoder feeding decentralised per-slice Q-heads, extended to federated training with differential privacy and evaluated under mid-episode disruption. Scaled from 3 to 19 gNBs, GAT-CTDE resists a training collapse that becomes total for a flattened-state single-agent baseline, converging, across 64 independently-drawn seeds, on a residual 35.4% [25.5%, 45.8%] collapse rate. Federating the architecture costs nothing measurable in reward, and differential privacy degrades decision precision only past a real noise threshold. Under gNB dropout, demand spikes, and agent churn, its disruption cost accelerates with severity like every arm tested, with a coordination-specific penalty only at the mildest severity. On the metric that reflects the actual training objective, GAT-CTDE decisively beats an independent-learner ablation, though its edge over the single-agent baseline remains small and not statistically decisive at this sample size – an honest limit we report rather than obscure. These results were made trustworthy, not just favourable, by two safeguards this paper also contributes: a correctness-aware metric pair that caught a real training collapse the standard SLA-compliance metric scored no worse than correct behaviour, and a reproduction-versus-replication discipline that caught and retracted one architectural overclaim under disruption before publication. A live single-gNB anchor confirms the architecture's decision logic transfers to real traffic unchanged.

**Keywords:** Network Slicing, Slice Admission Control, Multi-Agent Reinforcement Learning, Graph Attention Networks, Federated Learning, Open RAN

# 1 Introduction

Network slicing lets a single physical radio access network serve heterogeneous service classes – Ultra-Reliable Low-Latency Communication (URLLC), enhanced Mobile Broadband (eMBB), and massive Machine-Type Communication (mMTC) – under distinct per-slice Service Level Agreements (SLAs). Our earlier work [7, 8] established that a Deep Q-Network (DQN) admission controller can learn to enforce these SLAs under a single gNodeB (gNB), validated on a live OpenAirInterface (OAI) testbed. A real multi-cell deployment does not admit requests at one gNB in isolation: neighbouring cells share backhaul, spectrum planning, and a cluster-wide load-balance objective, making coordinated admission a genuinely multi-agent problem – does a shared, graph-structured representation of the cluster's state help a set of per-gNB agents make better joint decisions than a single agent using the flattened joint state, or independent per-gNB agents with no shared representation at all?

This paper's contribution is GAT-CTDE, a centralised-training, decentralised-execution multi-agent architecture built on a shared Graph Attention Network [17] encoder over the gNB topology feeding decentralised per-slice Q-heads, and the evidence for how it scales, coordinates, and degrades under realistic operating conditions. Extended to federated training with differential privacy and evaluated under mid-episode gNB dropout, demand spikes, and agent churn, we scale the architecture to cluster sizes $N \in \{7, 19\}$ under ring and hex-grid adjacency and find it resists a training collapse that becomes total for a flattened-state single-agent baseline, converging on a residual 35.4% collapse rate across 64 independently-drawn seeds rather than either of two disagreeing small-sample estimates. On the metric that reflects the actual training objective, the architecture decisively beats an independent-learner ablation that isolates the same coordination contribution, though its edge over the single-agent baseline specifically is small and not statistically decisive at this sample size – an honest limit we report rather than smooth over to fit a cleaner architecture story. Federating the architecture costs nothing measurable in reward, and differential privacy degrades decision precision only past a real threshold rather than a smooth trade-off; under disruption, its cost accelerates with severity like every arm tested, with a coordination-specific penalty at only the mildest severity tested.

Reporting these results honestly required two methodological safeguards we also contribute, since neither is standard practice in this literature. First, the standard episode-level SLA-compliance metric can score a degenerate, always-accept policy no worse than genuinely differentiated decision-making, because blocking a low-priority slice under congestion – the reward-optimal action – costs compliance without ever recovering it; we introduce a correctness-aware metric pair (mean reward per step and block precision, both computed from data every run already logs) that exposes this gap, and use it to diagnose and fix a real training collapse our first encoder suffered on all 30 campaign seeds. Second, we establish a reproduction-versus-replication discipline – checking every headline result against an independent seed sample, not only reproducing the same seeds – which caught and retracted a real architectural overclaim under mid-episode disruption before it could be reported as a finding. We finally anchor the architecture's single-gNB baseline against a real OAI testbed, confirming its offline-trained decision logic transfers to live traffic unchanged.

## 2 Related Work

Our own systematic review [9] screened 6,286 records down to 67 core 2024–2026 studies on machine-learning-driven, user-centric distributed network slicing, organised around three enabling pillars – topology-aware graph representation, distributed multi-agent control, and privacy-preserving federation – and found individual pillars well developed but rarely combined under realistic constraints, and never with a validated live deployment. None of the works we reviewed, there or here, checks whether its own evaluation metric can mask a degenerate policy, or replicates a headline finding on an independent seed sample – the gap this paper fills.

The canonical result behind our own encoder choice is Shao *et al.* [14], who show a graph attention encoder strengthens inter-agent cooperation over plain DQN/A2C baselines in dense-cellular slicing. Closest to our admission-control setting, Ahmadi *et al.* [2] generalise coordinated multi-agent slicing across unseen topologies via graph attention – a topology-generalisation question distinct from, but complementary to, the evaluation-integrity question we ask on a fixed topology (we test the complementary direction directly in Section 7: whether correctness-aware metrics remain reliable as cluster size and adjacency sparsity change, rather than whether a policy generalises to an unseen topology). Sulaiman *et al.* [15] is the closest prior art to our own problem, multi-agent DRL jointly solving slicing and admission control across the substrate network; neither that work nor its topology-generalisation follow-up reports whether its own reward-optimal policy could be mistaken for a degenerate one, or checks a headline result against an independent seed sample. Tashman and Cherkaoui [16] and Fatehi *et al.* [5] both study robustness to a disruptive condition, as our disruption campaign (Section 6.3) does; neither checks whether the metric reporting that robustness is itself trustworthy, or replicates a robustness finding on an independent sample – exactly the gap our own disruption campaign's independent-seed replication closes, and not hypothetically: it is what overturned our own initial churn-immunity reading.

On the federated side, Zhang *et al.* [19] coordinate independent O-RAN xApps via federated DRL, the closest precedent to our own periodic- synchronisation motivation, but without a differential-privacy step or a correctness-aware metric to check whether federating changed decision quality. Yasin *et al.* [18] combine federated edge learning with differential privacy to secure the O-RAN RIC – the closest FL+DP precedent – but target a different threat model (securing the controller) rather than characterising a control policy's own privacy-utility trade-off, which our $\sigma$-sweep (Section 6.2) reports. A small further set of works begins to combine two of these three pillars [3, 6, 12, 13], but, like every work surveyed here, evaluates exclusively on its own reported outcome metric, computed once, on one seed or run. This is the gap this paper fills: not a new point in the GAT/MARL/FL design space, but a demonstration – on GAT-CTDE, extended to federation, privacy, and disruption – that checking a metric's own trustworthiness and replicating a finding on an independent sample are both necessary, and that skipping either changes what a reader is entitled to believe the resulting numbers mean.

# 3 System Model and Problem Formulation

We consider a cluster of $N=3$ gNBs, each serving the same three slices (URLLC, eMBB, mMTC) under a shared per-gNB Physical Resource Block (PRB) budget $B$. An admission decision is realised as a PRB-ceiling adjustment (`slicing_control_m` min/max ratio), not a literal per-flow accept/reject, since the real OAI xApp control API only exposes ratio ceilings. No real multi-gNB physical topology is available to us, so we take the least assumption-laden choice and treat the cluster as fully connected, noting this explicitly as a modelling choice rather than a measured fact. At each control step, every gNB has zero or more pending admission requests, each tagged with a slice identity, and an agent decides accept/reject for each independently, conditioned on the current cluster state; the discrete action space matches our single-gNB work exactly (`action_dim=2`), so any coordination benefit found here is attributable to the architecture, not a richer action space.

The reward combines a per-slice SLA-compliance term with a cluster-wide load-balance penalty, weighted by a per-slice `priority_weight` (URLLC highest, mMTC lowest) and a congestion coefficient tuned, in prior work, so that the reward-optimal policy under congestion is *differentiated shedding*: continue accepting URLLC, mostly accept eMBB, and reject mMTC – confirmed directly by Q-value inspection on a working baseline policy (Section 5), not assumed. Every result in this paper except Section 8's bounded live anchor is obtained in an offline, closed-loop simulation environment deliberately oversubscribed to roughly 110% of one gNB's PRB budget, reusing our existing multi-gNB configuration unmodified. A separate investigation found this offline environment's demand process does not reproduce the live testbed's SLA-margin distribution closely enough to support a live-rank prediction claim, even after recalibrating the demand model's temporal structure; we therefore use it explicitly as a *stress environment* for a contention regime our live rig does not reach, not as a claim that any offline ranking would reproduce live – Section 8 tests this directly rather than only arguing for it. Multi-gNB claims in this paper remain offline-only throughout; no rig with more than one physical gNB exists to test them against.

# 4 GAT-CTDE Method

## 4.1 Shared Encoder, Decentralised Q-Heads

Fig. 1(a) shows the architecture. At every control step, each gNB's own per-slice [PRB-used-ratio, congestion-level, queue-length-norm] features form that node's row in a joint node-feature matrix. A two-layer, four-head GAT [17], implemented directly (the small graph sizes here make a dense, masked-attention implementation as fast as a sparse one), produces one embedding per gNB. Centralisation comes from this

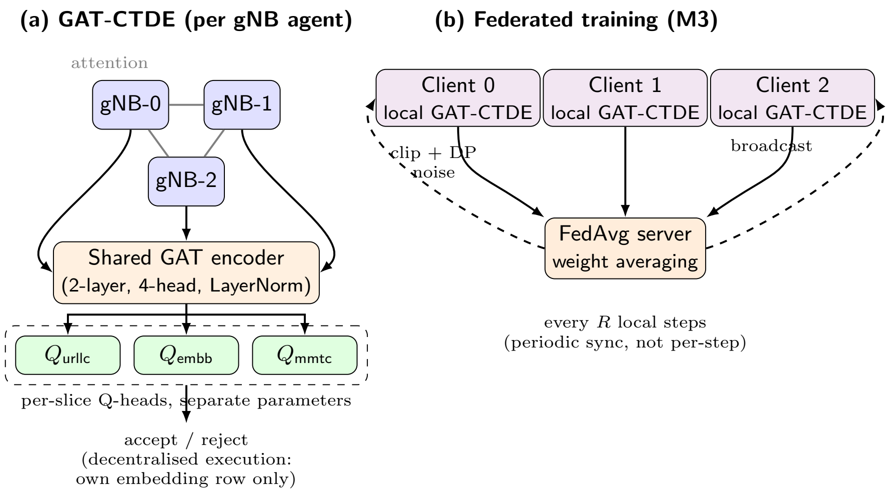


**Fig. 1** GAT-CTDE architecture. (a) A shared Graph Attention encoder over the fully-connected 3-gNB topology produces one embedding per node; per-slice Q-heads, each with its own parameters, consume that node's own embedding row at action-selection time (decentralised execution). (b) The federated training variant: each gNB is a client with a private copy of the same network, synchronised via periodic FedAvg with an optional per-client DP-SGD clip-and-noise step.

encoder: its parameters are updated by every agent's loss jointly, so it sees the full cluster's state during training. Decentralised execution comes from what happens next: at action-selection time, each agent consumes only its own node's embedding row to decide its own pending requests.

An early version fed each embedding, concatenated with a one-hot slice context, into a single shared Q-head MLP. Under this design, mmTC's true reward-optimal margin under congestion is small by the reward's own calibration, while URLLC's much larger priority weight produces proportionally larger gradients; because all three slices update the *same* shared parameters, URLLC's abundant, large-magnitude gradients overwrite mmTC's much smaller true signal before it registers – a standard shared-parameter/class-imbalance failure mode. We replace the single shared head with *one small MLP per slice*, selected by the context one-hot where the head is applied (Fig. 1(a)): each slice's decision boundary now has its own parameters, so one slice's gradients can no longer directly overwrite another's.

Algorithm 1 states the training loop: the encoder is updated from every request in a sampled minibatch regardless of gNB or slice (centralised training), while action selection at rollout time consults only the acting agent's own embedding row (decentralised execution). Every pending request in a step is its own independent temporal-difference sample, never summed or averaged across a step before the loss is computed, matching our existing single-agent DQN's own granularity.

## 4.2 Federated Training with Differential Privacy

Fig. 1(b) shows the federated variant: each gNB becomes an FL client with a private copy of the same network, training only on its own requests – transitions are

**Algorithm 1** GAT-CTDE: centralised training, decentralised execution

**Require:** shared GAT encoder $f_\theta$, per-slice Q-heads $\{Q_{\phi_k}\}_{k=1}^{K}$, target networks $f_{\theta^-}$, $\{Q_{\phi_k^-}\}$, replay buffer $\mathcal{D}$, fully-connected adjacency $A$, discount $\gamma$, target-sync period $C$ episodes

**for** episode $= 1, 2, \ldots$ **do**
  reset environment; observe joint node features $X_1 \in \mathbb{R}^{N\times d}$
  **for** step $t = 1, \ldots, T$ **do**
    $E_t \leftarrow f_\theta(X_t, A)$ {one embedding row per gNB; centralised: sees every node}
    **for** each pending request $r$ at gNB $i$, slice $k$ **do**
      $a_r \leftarrow \arg\max_a Q_{\phi_k}(E_t[i])[a]$ w.p. $1-\epsilon$, else uniform random {decentralised: own row $E_t[i]$ only}
    **end for**
    $X_{t+1}, \rho_t, \text{done} \leftarrow \textsc{env.step}(\{a_r\})$
    store $\big(X_t, X_{t+1}, \{(i, k, a_r)\}_r, \rho_t, \text{done}\big)$ in $\mathcal{D}$
    **if** $|\mathcal{D}| \geq$ warmup size **then**
      sample minibatch $B \subset \mathcal{D}$
      $E_B \leftarrow f_\theta(X_B, A)$;  $E'_B \leftarrow f_{\theta^-}(X'_B, A)$ {target net, no grad}
      **for** each request $(i, k, a) \in B$ **do**
        $q \leftarrow Q_{\phi_k}(E_B[i])[a]$
        $y \leftarrow \rho + \gamma \cdot (1-\text{done}) \cdot \max_{a'} Q_{\phi_k^-}(E'_B[i])[a']$
      **end for**
      $\mathcal{L} \leftarrow$ mean Huber loss over every request's $(q, y)$ pair {per-request, never summed per step}
      update $\theta$, $\{\phi_k\}$ by $\nabla\mathcal{L}$; clip grad-norm to 1.0
    **end if**
  **end for**
  **if** episode mod $C = 0$ **then**
    $\theta^- \leftarrow \theta$;  $\phi_k^- \leftarrow \phi_k\ \forall k$
  **end if**
**end for**

partitioned by owning agent, so no client sees another client's raw admission data – while its GAT encoder still consumes the full joint node-feature snapshot every arm observes. Every $R$ local steps, the server averages all clients' online weights (FedAvg [11]) and broadcasts the result back. We also implement FedProx [10] (a proximal term pulling local weights toward the last broadcast global model), motivated by an implicit, seeded per-gNB load multiplier this project's own cluster-scaling investigation (Section 7) later revealed as genuine, uncontrolled client heterogeneity: swept at $\mu \in \{0.01, 0.1, 1.0\}$ under both a homogeneous-load control and this heterogeneity, FedProx earns no measurable dividend at any tested strength, traced to a concrete mechanism – real per-round client drift within `local_steps_per_round` averages only 0.03% of the TD loss, too small for the proximal term to matter – and a follow-up at $10\times$ the round length lets real drift grow nearly 50-fold (to a mean 1.44%) yet still

**Algorithm 2** Federated GAT-CTDE round with DP-SGD

**Require:** $N$ clients, each with local network $(\theta_i, \{\phi_{i,k}\})$; global snapshot $\bar{\theta}$; local steps per round $R$; clip norm $C_{\mathrm{dp}}$; noise multiplier $\sigma$; FedProx weight $\mu$ ($= 0$ unless noted)

**for** round $= 1, 2, \ldots$ **do**
  **for** local step $= 1, \ldots, R$, each client $i$ independently **do**
    sample minibatch $B_i$ of $i$'s own requests from $i$'s local buffer
    $E \leftarrow f_{\theta_i}(X_{B_i}, A)$ {full joint state; client's OWN local weights only}
    $\mathcal{L}_i \leftarrow$ per-request Huber TD loss (Alg. 1, lines 11–13) $+ \frac{\mu}{2}\|\theta_i - \bar{\theta}\|^2$ {FedProx term}
    $g_i \leftarrow \nabla_{\theta_i,\phi_i}\mathcal{L}_i$
    **if** $\sigma > 0$ **then**
      $g_i \leftarrow \mathrm{clip}(g_i,\, C_{\mathrm{dp}})$; $g_i \leftarrow g_i + \mathcal{N}(0,\, (\sigma C_{\mathrm{dp}})^2 I)$ {DP-SGD [1]}
    **else**
      $g_i \leftarrow \mathrm{clip}(g_i,\, C_{\mathrm{dp}})$ {same clip norm, no noise – isolates the privacy cost}
    **end if**
    $(\theta_i, \phi_i) \leftarrow (\theta_i, \phi_i) - \eta\, g_i$
  **end for**
  $\bar{\theta}, \bar{\phi} \leftarrow \frac{1}{N}\sum_{i=1}^{N}(\theta_i, \phi_i)$ {FedAvg [11], equal client weights}
  **for** each client $i = 1, \ldots, N$ **do**
    $(\theta_i, \phi_i) \leftarrow (\bar{\theta}, \bar{\phi})$ {broadcast}
  **end for**
**end for**

produces eval-time behaviour bit-identical to FedAvg, ruling out round length as the limiting factor rather than leaving it untested.

Differential privacy is applied as a standard DP-SGD [1] step: every client's gradient is clipped to a fixed norm and, when enabled, perturbed with Gaussian noise scaled by a noise multiplier $\sigma$, before that client's own optimiser step. We report $\sigma$ as the primary privacy parameter, alongside a formal $(\varepsilon, \delta)$ bound computed via zero-concentrated differential privacy (zCDP) [4] composition of the Gaussian mechanism ($\rho = k/(2\sigma^2)$ over $k$ per-client releases, $\varepsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}$) – a conservative, non-subsampled bound, reported alongside $\sigma$ rather than in place of it. Algorithm 2 states one federated round precisely; clipping is applied identically whether or not noise is added, so a $\sigma$=0 run and a $\sigma$>0 run differ only in whether noise is injected – the isolation the privacy-cost comparison (Section 6.2) depends on.

## 4.3 Experimental Setup

Both the centralised and federated campaigns use 300 training and 50 evaluation episodes per seed. The centralised campaign uses a fixed 30-seed list (seeds 900–929) with three arms: GAT-CTDE; an independent-DQN ablation (one per-gNB DQN instance, no parameter sharing, no topology information) that isolates the GAT/centralised- training contribution; and a single-agent DQN baseline run against the flattened joint state. All three arms use matched hyperparameters (discount factor

$\gamma = 0.95$, per-episode epsilon decay). The federated campaign uses the first 10 of these 30 seeds and a five-level noise-multiplier sweep $\sigma \in \{0, 0.5, 1, 2, 4\}$, so its $\sigma$=0 level is directly comparable to the centralised campaign's same 10 seeds. All confidence intervals are 95% bootstrap percentile intervals (10,000 resamples); paired comparisons use the Wilcoxon signed-rank test.

# 5 Evaluation-Integrity Findings: A Training Collapse, Diagnosed and Fixed

While generating a privacy-utility sweep for the federated variant (Section 6.2), noise levels that should have produced genuinely different, stochastically-trained checkpoints instead produced bit-identical per-seed evaluation compliance. Tracing this back, we found that our first GAT-CTDE encoder's eval policy blocked zero requests across every evaluation episode, on all 30 campaign seeds – an always-accept policy is a functional no-op, so any two checkpoints that reach it produce identical trajectories regardless of their internal weights. We confirmed, rather than assumed, that this was a genuine failure and not a legitimate optimum: a direct Q-value probe on the collapsed checkpoint showed $Q(\text{accept}) - Q(\text{reject})$ positive and large for every slice, including mmTC, while the independent-DQN ablation (Section 6.1), which sees identical raw features through no encoder at all, does not exhibit this collapse – isolating the encoder, not the reward or training loop, as the fault.

Comparing the encoder's own embedding for a synthetic idle state against a congested one on the collapsed checkpoint: the raw input difference has $\ell_2$ norm 2.55 and the resulting embedding difference has norm 19.6 – an apparently large, congestion-sensitive response – but the cosine similarity between the two embeddings is 0.99997: the encoder was encoding congestion almost entirely as embedding *magnitude*, with no mechanism preventing this and no rotation of direction. Since the downstream Q-head is roughly linear in its input, larger magnitude pushes $Q(\text{accept})$ up faster than $Q(\text{reject})$ as congestion increases – backwards from the reward's intent.

We add a per-layer `LayerNorm` to the encoder, constraining every node's embedding to zero-mean, unit-variance regardless of input magnitude – a real, verified improvement (collapse drops from 30/30 to 27/30 seeds) but not a complete fix. We then apply the per-slice Q-heads of Section 4.1, which reduces collapse further, to 9/30 seeds (Fig. 2); a further variant disabling LayerNorm's learnable affine transform made results *worse* (0/3 pilot seeds differentiated versus 1/3 with it enabled) and was reverted – not every plausible fix helped, and we report the one that did not.

A separate, unrelated bug was caught while building an evaluation-time disruption harness: our logging utility opens its output file in append mode and never truncates it, and GAT-CTDE was retrained in place three times while diagnosing the collapse above, each time writing evaluation results to the same per-seed path – every one of the 30 seeds' evaluation logs had silently accumulated all three runs' episodes stacked together, confirmed directly by each log's own episode index resetting to 1 twice. Every GAT-CTDE number reported through an earlier draft of this paper, and every M3 result reusing GAT-CTDE's centralised evaluation as a reference point, was therefore a hidden three-way average across two architecturally-obsolete runs and the true final

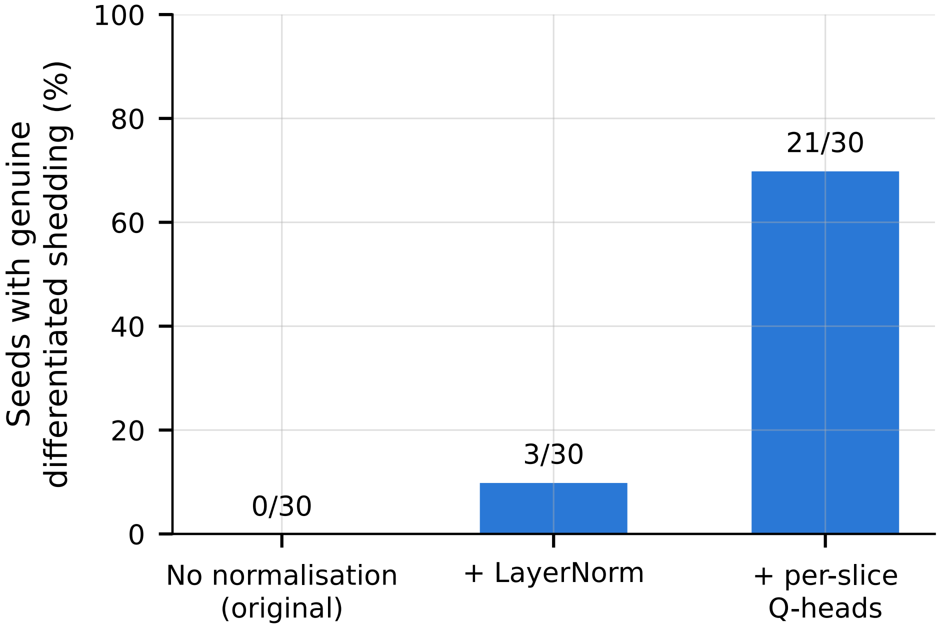


**Fig. 2** Fraction of the 30 campaign seeds reaching genuine, correctly-targeted differentiated shedding (mMTC-only blocking under congestion), across the original unnormalised encoder and the two fixes described in Section 5.

one. The saved model checkpoints themselves were unaffected (a different save routine overwrites rather than appends, verified by reproducing one seed's final-block statistics exactly from its checkpoint in isolation), so the fix required no retraining: we cleared every stale log, re-ran evaluation once against the already-trained checkpoints, and hardened the training script to clear its output path before any future run. All numbers in this paper reflect that corrected data.

The fixes above make the architecture more correct while making its `sla_compliance_all_slices` score *worse* (Section 6.1): blocking mmTC under congestion, the reward-optimal action, does not recover mmTC's own SLA margin in this stress regime, so every newly-correct decision costs compliance the metric never credits back. We therefore report two further metrics, both computed from data every arm already logs: **mean reward per step**, the actual training objective; and **block precision**, the fraction of blocked requests that target mmTC specifically, the only slice the reward calibration ever makes reject-optimal (undefined, and reported as such, for seeds that never block anything). Any admission-control evaluation where correct behaviour cannot recover an already-stressed slice's compliance should report a decision-level correctness metric alongside, not instead of, an outcome-level compliance one – the general lesson this specific collapse makes concrete.

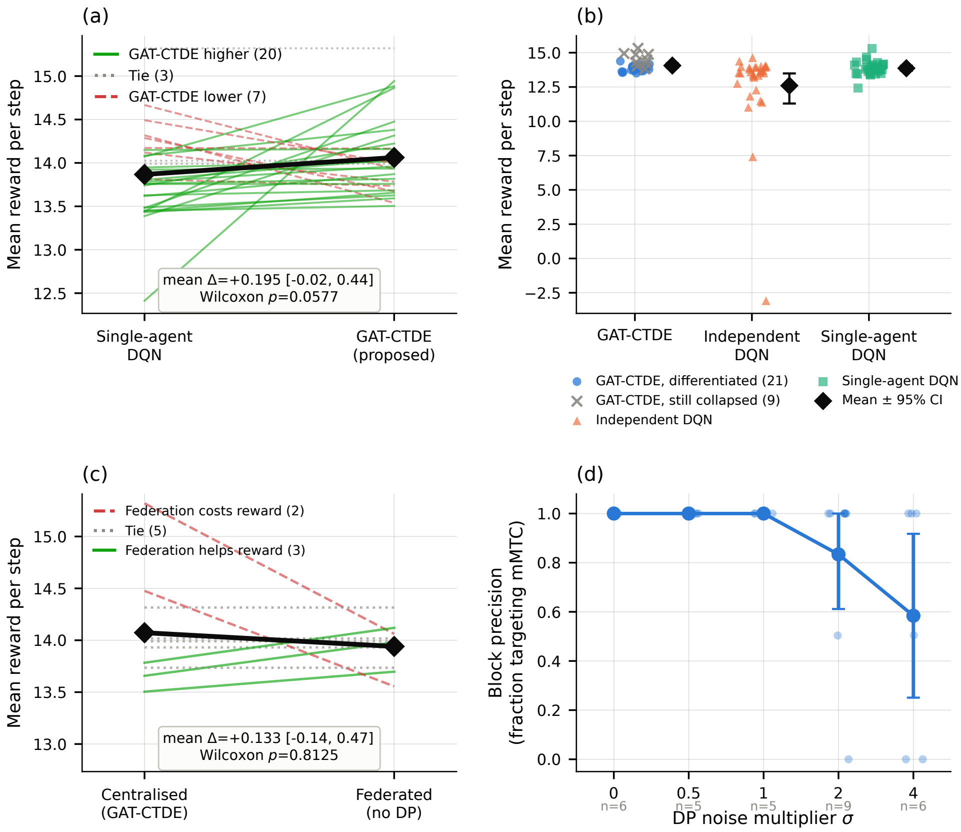


**Fig. 3** Multi-gNB results, per-seed evidence for the correctness-aware metrics (Section 5). (a) M2, 30 seeds: paired per-seed comparison against single-agent DQN on mean reward per step – each line is one seed, coloured by whether GAT-CTDE scored higher (green, 20), tied (grey, 3), or lower (red, 7) – the source of this figure's borderline $p = 0.0577$. (b) M2: full per-seed reward distribution for all three arms; GAT-CTDE is split into seeds that learned genuine differentiated shedding (blue, 21) versus seeds still stuck at the always-accept collapse (grey $\times$, 9) – shown without claiming the split predicts reward, since several still-collapsed seeds score among the arm's highest. (c) M3, same 10 seeds: paired per-seed federation cost on mean reward per step – centralised scores higher on 2 seeds, federated on 3, 5 ties, no measurable difference (Wilcoxon $p = 0.8125$). (d) M3: privacy-utility curve using block precision against the DP noise multiplier $\sigma$, individual seed values (light dots) behind the mean, with the seed count that ever blocked anything at each $\sigma$ printed below (precision is undefined otherwise) – perfect through $\sigma = 1.0$, dropping sharply by $\sigma = 4.0$.

**Table 1** M2 campaign results, 30 seeds/arm, 95% bootstrap CIs.

| Arm | Compliance | Mean reward/step | Block precision |
|---|---|---|---|
| GAT-CTDE | 0.151 [0.083, 0.225] | **14.062** [13.91, 14.23] | 0.952 |
| Indep. DQN | 0.049 [0.013, 0.093] | 12.601 [11.30, 13.49] | 0.901 |
| Single-agent | 0.115 [0.061, 0.176] | 13.867 [13.69, 14.05] | 0.987 |

# 6 Multi-gNB Results: Centralised, Federated, and Disruption

## 6.1 Centralised Campaign

Table 1 and Fig. 3(a)-(b) report the final, post-fix 30-seed campaign. On `sla_compliance_all_slices`, GAT-CTDE's mean is 0.151, below its own pre-fix number (0.353) since the metric penalises newly-correct mmTC-blocking decisions, and the paired comparison against single-agent DQN shows no clear direction: +0.036 (95% CI [−0.050, 0.126], Wilcoxon $p = 0.4549$, 11 wins / 9 ties / 10 losses). On mean reward per step, the metric that reflects the actual objective, GAT-CTDE beats the independent-DQN ablation decisively – +1.461 (95% CI [0.550, 2.782], $p < 0.0001$, 27/0/3), isolating a real contribution from the shared GAT encoder and centralised training – but its edge over single-agent DQN is small and not statistically decisive at this sample size: +0.195 (95% CI [−0.017, 0.442], $p = 0.0577$, 20/3/7). Its block precision (0.952) is high regardless: when GAT-CTDE blocks a request, it is almost always the reward-correct slice. 21 of the 30 seeds show genuine, correctly-targeted differentiated shedding, up from 3/30 under the LayerNorm-only fix and 0/30 originally – a real, substantial behavioural improvement that does not, at this sample size, translate into a decisive reward edge over the single-agent baseline specifically.

## 6.2 Federated Training and Differential Privacy

Comparing the centralised campaign against the federated variant with no DP noise, on the same 10 seeds: mean reward per step moves from 14.073 to 13.940, +0.133 (95% CI [−0.138, 0.470], Wilcoxon $p = 0.8125$) – no measurable cost to federating at this sample size, and both arms' block precision is perfect (1.000) on seeds that block anything at all. Sweeping the DP noise multiplier (Fig. 3(d)), block precision is perfect through $\sigma = 1.0$ then drops sharply – 0.834 at $\sigma = 2.0$, 0.584 at $\sigma = 4.0$ – a real threshold, not a smooth degradation, while mean reward per step stays in a comparatively narrow band across the whole sweep (12.45–14.19): DP noise erodes *which* slice gets blocked, not whether the policy tries to differentiate at all. The compliance-based version of this same sweep (not shown) is non-monotonic with sign flips between adjacent $\sigma$ levels – a second instance, independent of Section 5's architectural finding, of `sla_compliance_all_slices` failing to track the quantity we actually care about.

**Table 2** Training hyperparameters, all fixed across seeds.

| Hyperparameter | Value |
|---|---|
| Learning rate (GAT-CTDE) | $1\times10^{-4}$ |
| Learning rate (DQN baselines) | $1\times10^{-3}$ |
| Discount $\gamma$ | 0.95 |
| Epsilon: start $\rightarrow$ end (decay/episode) | 1.0 $\rightarrow$ 0.05 (0.985) |
| Target-network sync | every 10 episodes |
| Optimiser | Adam |
| Training / evaluation episodes | 300 / 50 |
| *GAT encoder (GAT-CTDE only)* | |
| Hidden / output dim | 16 / 16 |
| Attention heads | 4 |
| Encoder layers | 2 |
| Q-head hidden dim | 32 |
| *Federated + DP (federated GAT-CTDE only)* | |
| Local steps per round | 50 |
| Aggregator | FedAvg ($\mu$=0) |
| DP gradient clip norm | 1.0 |
| DP noise multiplier sweep $\sigma$ | $\{0, 0.5, 1, 2, 4\}$ |

## 6.3 Disruption Resilience

We evaluate every arm's already-trained, frozen checkpoint (no retraining) against three mid-episode disruptions injected at evaluation time: *gNB dropout* (one gNB's requests are forced to reject and its row in the joint feature matrix is zeroed, for a window); *demand spike* (a transient arrival-rate multiplier for a fixed window); and *agent churn* (one gNB's decisions are made by a fresh, never-trained copy of its own policy class instead of the loaded checkpoint, for a window) – a different, well-motivated failure mode from dropout (active-but-uncoordinated vs. absent) that applies to every genuinely multi-agent arm; single-agent DQN is excluded. Severity is the window's duration (10%/30%/60% of the episode) for dropout and churn, and the arrival-rate multiplier (2×/4×/8×) for spike. We normalise reward by each step's own pending-request count before averaging, since raw reward mechanically inflates with request volume regardless of decision quality.

Each disruption kind intercepts the frozen policy's per-step evaluation loop differently: dropout acts twice, corrupting the target gNB's input row and separately forcing that gNB's own requests to reject, so a policy cannot simply learn around the corrupted feature; agent churn substitutes a fresh, never-trained policy's actions for the target gNB only, leaving every other agent's decisions untouched; demand spike is entirely external to the policy, acting on the arrival process the environment synthesises for the following step. All three are evaluation-only perturbations – the policy itself is never updated.

Dropout costs every arm significantly at every severity (Wilcoxon $p \leq 0.03$, 9 or 10 of 10 seeds hurt in every case), and the cost accelerates with severity rather than growing proportionally: for GAT-CTDE, +0.165 at the 10% window, +0.601 at 30%, +1.710 at 60% – the same super-linear pattern holds for every arm. Agent churn shows

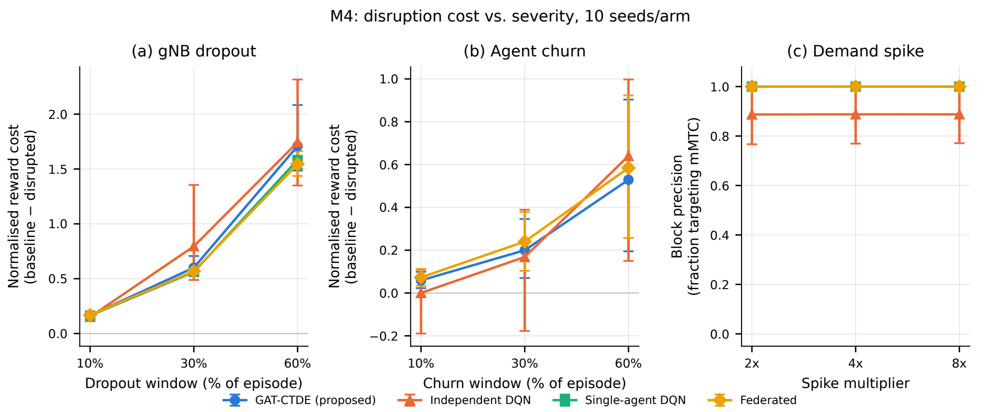


**Fig. 4** M4 disruption campaign, 10 seeds/arm. (a)-(b) Normalised reward cost (baseline − disrupted) vs. severity for dropout and agent churn – every arm, every severity, shows a real cost that accelerates with severity rather than growing linearly. (c) Block precision vs. demand-spike multiplier: flat and near-ceiling for every arm, unlike (a)-(b) – spike does not threshold on the metric that matters.

the same accelerating shape for GAT-CTDE (+0.059/+0.200/+0.529, $p = 0.0039$ throughout) and the federated arm (+0.073/+0.240/+0.585, $p = 0.0039$ throughout). Independent DQN's own churn cost needs both samples reported together to state honestly: on the seeds used throughout the rest of this paper (900–909), it is small and not conventionally significant (+0.000/+0.168/+0.640, $p = 0.084/0.106/0.065$) – which an earlier pass read as independent DQN being architecturally immune to churn, since it never attends to another agent's features. An independent-seed replication (seeds 1000–1009, disjoint from every other result in this paper) contradicts that reading directly: independent DQN's churn cost there is significant at every severity (+0.097/+0.657/+1.223, $p = 0.0020$ throughout) and *larger* than GAT-CTDE's or the federated arm's at matched severities – the opposite of immune. We do not conclude independent DQN is immune to churn: the direction was consistent across both samples (churn always hurts, never helps); what was wrong was treating a borderline, underpowered $n = 10$ result as a confirmed architectural property, and this section reports the corrected reading, not the retracted one, as this paper's discipline of checking every headline result against an independent seed sample – not the same seeds reproduced again – requires.

Demand spike shows no threshold at all on block precision, which stays flat and near-ceiling through the highest tested multiplier for every arm; a reward-based signal that spike *improves* per-step reward is real and reproduces across both seed samples for GAT-CTDE, independent DQN, and the federated arm ($p \leq 0.014$ in both), but traces to a fixed per-step violation penalty diluted by a larger request-volume denominator, not better decision-making – block precision, a volume-invariant ratio, is the metric that actually answers whether spike corrupts decision quality, and it says no. Single-agent DQN's version of this same signal is significant in the original sample ($p =$

0.0020) but not the replication ($p = 0.084$) – direction consistent, significance sample-dependent, reported as such.

Every dropout and churn result above was independently checked against the 1000–1009 sample and holds up in direction and approximate magnitude – e.g. GAT-CTDE's dropout costs there are +0.170/+0.552/+1.449, close to the 900–909 sample's +0.165/+0.601/+1.710. Every headline paired comparison in this paper was checked this way, recomputed from raw logs on both samples: GAT-CTDE's edge over independent DQN (Section 6.1) holds decisively on the replication sample too (+0.767, 95% CI [0.391, 1.233], $p < 0.0001$, versus the committed sample's +1.461 [0.550, 2.782]); its edge over single-agent DQN stays non-decisive on both (+0.445 [−0.049, 1.280], $p = 0.111$, versus +0.195 [−0.017, 0.442], $p = 0.058$); federation's reward cost stays statistically indistinguishable from zero on both, with the point estimate's sign flipping between samples (−0.080 [−0.241, 0.001] on replication versus +0.133 [−0.138, 0.470] committed) but neither ever significant. Of the seven headline paired comparisons checked this way across the whole paper, six show overlapping confidence intervals across samples; the one exception – independent DQN's churn cost – is the retraction above.

A formal paired Wilcoxon test on (GAT-CTDE's cost − independent DQN's cost), computed across the same 10 seeds for every (kind, severity) cell, reaches significance only once: dropout at the mildest severity (10% window), where GAT-CTDE's cost is significantly *higher* than independent DQN's (+0.0135, 95% CI [0.0053, 0.0236], Wilcoxon $p = 0.0059$, 9 losses / 0 ties / 1 win). Every other cell – dropout at 30%/60% and churn at all three severities – shows no significant difference ($p \geq 0.23$). We report the result as it is: real at one severity, absent everywhere else tested, not a general claim that coordination costs more under disruption.

# 7 Cluster-Size Scaling

We extend the cluster from $N = 3$ to $N \in \{7, 19\}$ under three adjacency structures (fully-connected, ring, hex-grid), scaling synthetic arrivals with $N$ to hold offered load per gNB roughly constant. GAT-CTDE is the only arm whose Q-head consumes this adjacency; independent DQN and single-agent DQN never read a graph structure and serve as a topology-invariance check. We normalise reward per gNB before averaging (the same fix Section 6.3 applies on the severity axis) and report block precision alongside, since `sla_compliance_all_slices` is OR-aggregated across gNBs and trends toward 0 as $N$ grows regardless of policy quality – a third instance of this paper's central evaluation-integrity finding, now on the cluster-size axis. The primary sample is 12 seeds (900–911) at $N = 19$, all three topologies, all three arms; an independent-seed replication (1000–1002, disjoint) checks generalisation.

At the full 12-seed sample, the paired per-gNB reward-margin comparison at $N = 19$ is not significant for any topology (fully-connected −0.175 [95% CI −0.528, +0.076], $p = 0.7422$; ring −0.704 [−1.879, +0.042], $p = 0.7422$; hex −0.695 [−1.873, +0.053], $p = 0.9102$) – mechanically explicable, not a red flag: single-agent DQN's always-accept collapse (below) collects the reward's service term on every request with no rejection cost, while GAT-CTDE pays a real cost every time it correctly rejects mmTC. This

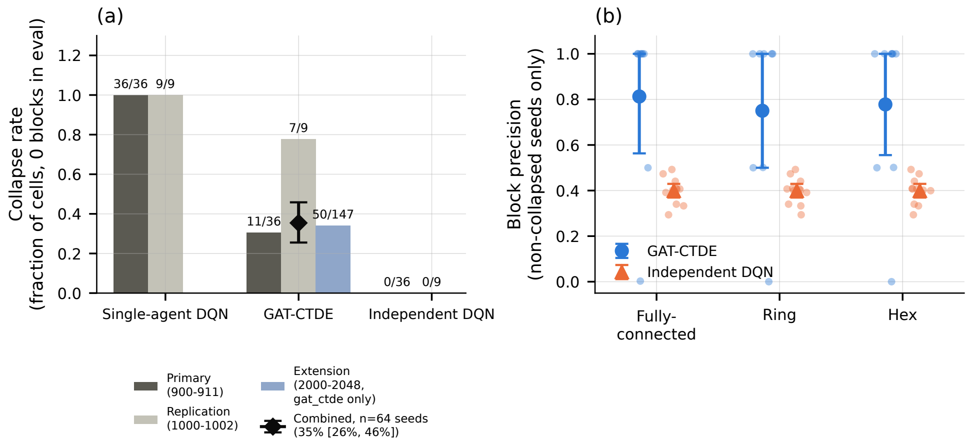


**Fig. 5** $N = 19$ cluster, per-seed evidence across independent seed samples. (a) Collapse rate (fraction of eval runs with zero blocks), pooled across all three topologies – single-agent DQN collapses totally in every sample; GAT-CTDE's rate varies sample-to-sample (primary 31%, replication 78%, a 49-seed extension targeted specifically at this question 34%), converging to a combined, seed-level-bootstrapped estimate of 35.4% [25.5%, 45.8%] across all 64 independent seeds (diamond marker, error bar). (b) Block precision by topology, primary sample, non-collapsed seeds only (single-agent DQN has none to show): GAT-CTDE's non-collapsed seeds are mostly near-ceiling but with a low-precision seed at every topology, while independent DQN sits in a narrow, mediocre band, identical across topologies as expected for an arm that never consumes adjacency.

paper's evidence for GAT-CTDE's advantage at $N = 19$ lives in collapse resistance and block precision, not the reward-margin metric.

Single-agent DQN collapses totally at $N = 19$: 12/12 primary-sample cells and 3/3 replication cells, every topology, zero exceptions (15/15 combined) – confirmed at the individual-decision level by a direct Q-value probe identical in method to Section 5's original diagnosis. GAT-CTDE collapses only partially, and independent DQN never fully collapses (0/36) but never cleanly differentiates either, sitting in a narrow, mediocre 0.29–0.49 block-precision band identical across all three topologies, as expected for an arm that never consumes adjacency. GAT-CTDE's own non-collapsed seeds show a topology-dependent secondary failure mode: most hold near-perfect precision, but at least one seed per topology drops to 0.0–0.5, including one seed showing perfect precision (1.000) at fully-connected and near-zero (0.000) at ring/hex on the identical seed, environment stream, and initial weights.

GAT-CTDE's own collapse rate initially disagreed sharply between the primary sample (31%) and the replication sample (78%); rather than report the range as open, we drew a third, independent sample targeted specifically at this question (49 seeds, disjoint from both prior samples). Pooling all three samples gives 68/192 cells collapsed, 35.4% (seed-level bootstrap 95% CI [25.5%, 45.8%] across the 64 independent seeds, since collapse status correlates within a seed across its three topologies) – GAT-CTDE's collapse rate at $N = 19$ is best understood as roughly 35% with substantial seed-to-seed variability, and none of the three samples was wrong: each was an honest reading of a genuinely wide distribution, and no single small sample should be trusted alone.

# 8 Live Single-gNB Anchor

Prior live-testbed work on this rig already found that the offline environment cannot predict live *ranking* (Spearman $\rho = 0.097$, $p = 0.855$, across six single-gNB checkpoints); we do not repeat that attempt, and instead ask a narrower, bounded question this rig can answer: does the offline-trained *decision logic* survive contact with real traffic at all. This rig has one physical gNB, so GAT-CTDE's multi-gNB coordination benefit and the federated/DP results are architecturally untestable on it; we trained a fresh single-agent-DQN checkpoint dimension-matched to the live evaluation path and ran it against a real OAI gNB, 3 UEs, and real per-slice UDP traffic, twice – once from a leftover backlog state, once from a verified-clean baseline.

Both live runs blocked only mmTC (46/32 and 45/32 requests, zero urllc or eMBB blocks in either), exactly matching this checkpoint's offline block precision of 1.000: the offline-trained differentiated-shedding decision transfers to live traffic essentially unchanged, replicated across two independent runs. Both runs' SLA-margin readings, by contrast, diverged from anything offline produces – eMBB's margin fell from +1.0 to −1,002,377.5 within a 10-minute run, recurring to the decimal in the independent second run – traced to a concrete unit mismatch, not left unexplained: the margin computation divides a per-slice backlog reading by a fixed calibration constant tuned for the offline environment's synthetic proxy, but live, the identically-named field is wired to the real MAC scheduler's transmit-buffer byte count, a physically real quantity two to three orders of magnitude larger in scale. The policy's own decision-facing input escapes this (a separate path clips the state-facing backlog term before it reaches the network), which is why the policy's decisions stayed sensible live even as this diagnostic margin exploded. Checking 28 seeds of an already-committed prior live campaign on this rig confirms the calibration constant is not simply "wrong": it reproduces a sensible number under normal conditions and only explodes once a real, independently-documented backlog failure regime is entered. Rather than leave the mismatch diagnosed-but- unpatched, we recalibrated using a Little's-Law byte-to-latency conversion against each slice's own latency budget – a physically meaningful quantity rather than an arbitrary rescaling constant – which brings the catastrophic case (≈10.02MB backlog) to −809 to −1,187 (36–53 real seconds of queueing delay against a 45ms budget) rather than −1,002,377.5: three orders of magnitude smaller and directly interpretable, without reopening the rank-prediction question already closed above.

# 9 Conclusion

We set out to test whether a standard evaluation metric can be trusted in multi-agent, privacy-constrained, disruption-prone slice admission control, and found it cannot go unchecked. A standard SLA-compliance metric, an unguarded append-only log, and an unseeded weight-initialisation path together hid three distinct failures – a training collapse the compliance metric scored no worse than correct behaviour, a hidden eval-data contamination bug, and a same-seed-only reproducibility gap – none of which a standard pipeline without this paper's correctness-aware metrics and reproduction-replication discipline would have caught.

On GAT-CTDE, the multi-agent architecture that served as this evaluation's substrate: on the correctness-aware reward metric it delivers a decisive, statistically significant improvement over an independent-learner ablation, though its edge over our existing single-agent baseline specifically is small and not decisive at this sample size – an honest, non-inflated architectural finding, not the headline result this paper argues evaluation practice should not be built around alone.

Extending to federated training with differential privacy, federation adds no measurable reward cost, and privacy is costly only past a real threshold in decision precision, not a smooth trade-off; FedProx earns no measurable dividend under genuine per-gNB load heterogeneity at any tested strength, traced to real per-round client drift being too small relative to the loss scale to matter, even at ten times the round length. Under mid-episode gNB dropout, demand spikes, and agent churn, dropout and churn produce a real, accelerating cost with severity, while demand spikes do not threshold at all on the metric that matters – and cross-checking every result against an independent seed sample, not just the same seeds reproduced again, caught one architectural claim (that independent DQN is immune to agent churn) that a single sample had made look confirmed but was not, retracted here rather than reported.

Scaling the cluster to 19 gNBs, single-agent DQN's collapse becomes total and GAT-CTDE's remains only partial, a real cluster-size-specific effect; GAT-CTDE's own collapse rate, read at two disagreeing values across two small samples (31% vs. 78%), was resolved rather than left open with a third, independently-drawn sample, converging on 35.4% [25.5%, 45.8%] across 64 seeds. Finally, the offline-trained admission-control decision logic transferred to a real single-gNB testbed essentially unchanged, while its SLA-margin metric diverged by orders of magnitude for reasons traced to a concrete unit mismatch rather than left unexplained.

Two limits bound every claim in this paper. First, every multi-gNB, federated, and disruption result is offline-only: this rig has one physical gNB, so whether these rankings would survive contact with a live, multi-gNB testbed remains untested, not just under-tested. Second, this paper's federated scenario is a synthetic privacy setting – one operator's cluster, not genuine cross-organisation data separation – so we lead with the operational benefit of periodic, not per-step, synchronisation for disaggregated RAN Intelligent Controllers rather than a privacy motivation this scale cannot substantiate. Together, these results argue that admission-control evaluation in multi-agent, privacy-constrained, disruption-prone settings needs metrics tied to the actual training objective, not only to a downstream SLA proxy that can reward the wrong policy, and that architectural claims need more than one seed sample before they are trusted.

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## Declarations

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- **Ethics approval and consent to participate:** Not applicable – this study involves no human participants, animal subjects, or identifiable patient data; all reported results are either offline simulation campaigns or measurements from a laboratory radio testbed.
- **Consent for publication:** Not applicable – this manuscript contains no personal data or identifying information about any individual.
- **Data availability:** The experiment logs, checkpoints, and campaign result files underlying this paper's findings are available from the corresponding author on reasonable request.
- **Materials availability:** Not applicable – this study involves no physical or biological materials.
- **Code availability:** The framework and experiment scripts used to produce this paper's results are available from the corresponding author on reasonable request.
- **Author contribution:**

  **TODO(MANOJ):** State each author's individual contribution (e.g., conceptualisation, methodology, software, investigation, writing, supervision), per CRediT taxonomy or equivalent.

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